

Digital Logic & Number Systems Question Bank

1. Number Systems

1. Convert $(1101)_2$, $(1101)_2$ to decimal.
 2. Convert $(347)_8$, $(347)_8$ to decimal.
 3. Convert $(2A5)_{16}$, $(2A5)_{16}$ to binary.
 4. Convert $(101101.101)_2$, $(101101.101)_2$ to octal.
 5. Convert $(375)_{10}$, $(375)_{10}$ to hexadecimal.
 6. Perform $(1011)_2 + (1101)_2$, $(1011)_2 + (1101)_2$ using binary addition.
 7. Subtract $(11001)_2 - (10110)_2$, $(11001)_2 - (10110)_2$ using the 2's complement method.
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2. Base Arithmetic

1. Multiply $(101)_2 \times (11)_2$, $(101)_2 \times (11)_2$ and express the result in binary and decimal.
 2. Add $(235)_8 + (457)_8$, $(235)_8 + (457)_8$ and express the result in octal.
 3. Subtract $(A6)_{16} - (3B)_{16}$, $(A6)_{16} - (3B)_{16}$ and express the result in hexadecimal.
 4. Multiply $(24)_{10} \times (15)_{10}$, $(24)_{10} \times (15)_{10}$ and convert the result to binary.
 5. Perform $(1A)_{16} + (1110)_2$, $(1A)_{16} + (1110)_2$. Express the answer in hexadecimal.
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3. Logic Gates

1. Draw the truth tables for **AND**, **OR**, **NOT** gates.
 2. Construct **XOR gate using only NAND gates**. Show truth table.
 3. Prove that **NAND is a universal gate** by constructing AND and OR using NAND.
 4. Design a circuit using logic gates that gives output **1 only when exactly two inputs out of A, B, C are 1**.
 5. Write the truth table for $(A+B)(A'+C)(A+B)(A'+C)$.
 6. Simplify the Boolean expression: $(A+B)(A'+C)(B+C)(A+B)(A'+C)(B+C)$.
 7. Design the truth table for a **Half-Adder** and provide Boolean expressions for **Sum** and **Carry**.
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4. Truth Tables & Expressions

1. Write the truth table for the expression: $F = A \cdot B + A' \cdot C = A \cdot B + A' \cdot C$.
2. Simplify using Boolean algebra: $X + X'YX + X'Y$.
3. Reduce the expression: $F = (A+B)(A+C)(B+C)$ $F = (A+B)(A+C)(B+C)$.
4. Construct the truth table for $F = (A+B')(B+C)$ $F = (A+B')(B+C)$.
5. Simplify $F(A,B,C) = \Sigma(0,1,2,5,6,7)$ $F(A,B,C) = \Sigma(0,1,2,5,6,7)$ using **Karnaugh Map**.
6. Prove **De Morgan's Laws** using truth tables.
7. Write the **SOP (Sum of Products)** form for the truth table where $F=1$ for minterms 1, 3, 5, 7.